MILITARY SPECIFICATION SHEET

NUT, SELF-LOCKING, ASSEMBLED WASHER, A-286, 160 KSI Ftu 450°F, SPLINE WRENCHING ELEMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the nut described herein shall consist of this document and the latest issue of MIL-N-85729.

1. Design selection: These nuts are designed for use with expandable sleeve bolts conforming to MIL-B-85667.

2. Application criteria: The nuts shall be used in accordance with the limitations of MS33588.

3. Tooling: Per MIL-W-8982

4. General requirements:

4.1 Material.

4 1.1 Nut· Cres (A-286) per AMS5737, and capable of 160 KSI minimum ultimate tensile strength

4.1.2 Washer: Cres (A-286) per AMS5525

4.2 Heat Treatment:

4.2.1 Nut: Heat treat to 160 KSI min per MIL-H-6875

4.2.2 Washer: Heat treat to 140 KSI min per MIL-H-6875.

4 3 Surface Treatment:

4.3.1 Nut: Passivate per QQ-P-35.

4 3 2 Washer. Passivate per QQ-P-35

4.4 Lubricant:

4.4.1 Nut:

4.4.1.1 Dry Film lubricate per MIL-L-8937

AMSC N/A FSC 5310

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4.4.1 2 Cetyl alcohol, Type I, Grade A lubrication per MIL-L-87132.

4.4.2 Washer:

4.4.2.1 Dry film lubricate per MIL-L-8937.

4.4.2.2 Cetyl alcohol, Type I, Grade A lubrication per MIL-L-87132

4.5 Workmanship:

4 5 1 Dimensions are in inches.

4.5.1.1 Dimensions apply before lubrication.

4.5.2 Tolerances: As specified in Figure 1 and Table I.

4.6 Bearing Surface: Bearing surface shall be normal with pitch diameter of thread within the Y values specified, see table I.

4.7 Concentricity: N diameter shall be concentric to pitch diameter of thread with X values T.I.R., see table I.

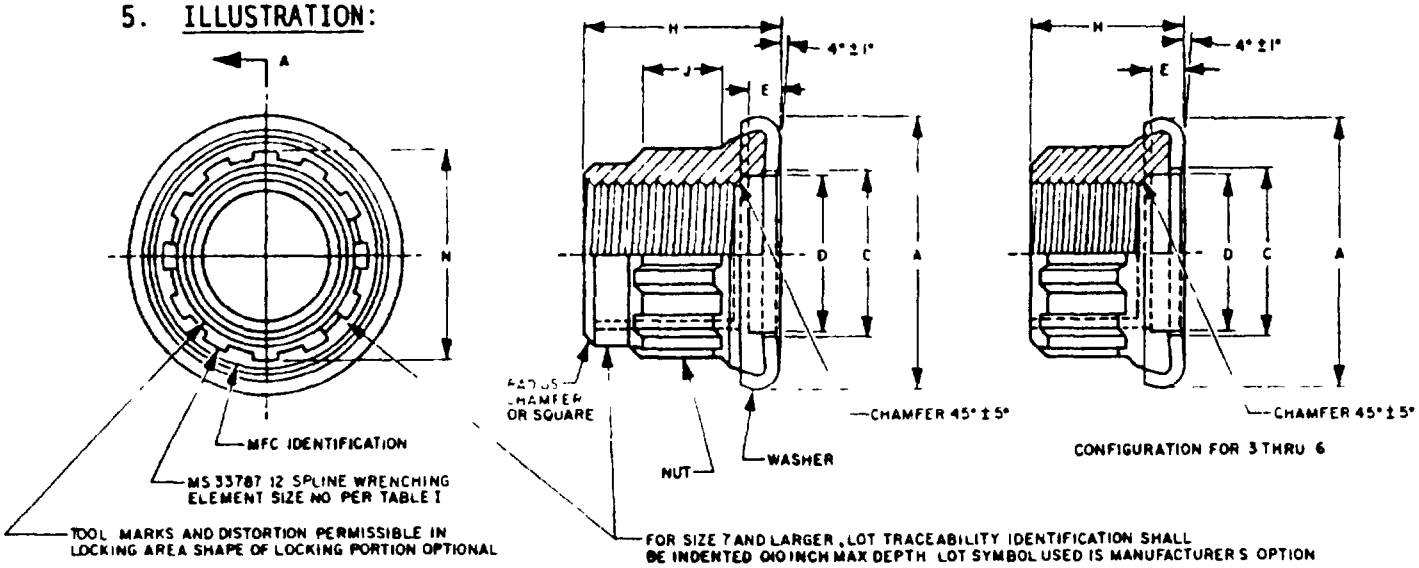
4.8 Example of military part number.

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- 4 — Thread. 0.250 - 28 UNJF-3B
- \- — Lubricant: " - " denotes cetyl alcohol lubricant in accordance with MIL-L-87132. "L" denotes dry film lubricant.
- M85729/1 — Basic Specification Sheet Number

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5. ILLUSTRATION:



6. TABULATION:

TABLE I Nut Dimensions and Strengths

SIZE NUMBER	THREAD	MS33787 (12 SPLINE) ELEMENT NUMBER	A MAX	C ±.005	D ±.005	E +.010 −.005
3	.1900-32UNJF-3B	8	.460	.288	.282	.077
4	.2500-28UNJF-3B	10	.560	.353	.347	.077
5	.3125-24UNJF-3B	12	.660	418	412	.077
6	.3750-24UNJF-3B	14	.760	.485	.479	.077
7	.4375-20UNJF-3B	18	.860	552	546	.077
8	.5000-20UNJF-3B	20	1.000	.636	.621	.077
9	.5625-18UNJF-3B	22	1.100	.765	.750	.087
10	.6250-18UNJF-3B	24	1.310	.840	.825	.087
12	.7500-16UNJF-3B	30	1.480	.987	.972	.092
14	.8750-14UNJF-3B	34	1.690	1.111	1.096	.097
16	1.0000-12UNJF-3B	38	1.834	1.253	1.238	.097
18	1.1250-12UNJF-3B	44	2.100	1.425	1.410	.097
20	1.2500-12UNJF-3B	48	2.250	1.500	1.485	.097
22	1.3750-12UNJF-3B	52	2.480	1.650	1.635	.097
24	1.5000-12UNJF-3B	56	2.690	1.780	1.765	.097

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TABLE I. Nut Dimensions and Strengths

SIZE NUMBER	H		J	N	X	Y	ULTIMATE AXIAL TENSILE STRENGTH LB MIN (a)
	MAX	MIN	MIN	MAX			
3	.280	260	.080	308	.003	.004	3,620
4	.332	.312	100	.381	.003	.004	6,480
5	400	380	120	.456	003	.005	10,240
6	.478	.458	.152	.531	003	.006	15,220
7	.525	505	.168	679	003	.008	20,650
8	.603	.583	.190	.753	003	.008	27,500
9	672	652	230	828	004	009	34,800
10	734	.714	.250	.901	.004	.010	43,600
12	879	859	310	1 124	.004	.012	63,200
14	1 012	992	.350	1.270	.005	.014	86,720
16	1.140	1 120	.400	1.419	.006	016	114,800
18	1 337	1 317	.450	1.641	007	019	144,180
20	1.524	1.504	500	1.788	.007	.021	179,780
22	1.659	1.639	550	1 937	.008	.023	218,940
24	1.847	1 827	600	2 085	008	.023	263,740

(a) Axial tensile strength determined from formula Wa = FtuA. Where A is the cross sectional area in square inches, based on the maximum pitch diameter of bolt thread, Ftu is 160 KSI and Wa is the axial tensile strength in pounds

7.0 Administrative and contractual provisions:

7 1 For design feature purposes this document takes precedence over procurement documents referenced herein.

7.2 Referenced documents shall be of the issue in effect on date of invitation for bid.

Custodians:
Army - AV
Navy - AS
Air Force -99

Review activities:
DLA-IS
Air Force - 82

Preparing activity:
Navy - AS
(Project No. 5310-1424)

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